Project II

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1. Project II

Project II

Faculty:	Information Technology
Module Code:	ITPNA2-12
Module Name:	Python Network Programming
Content Writer:	Mr M Nduwamungu
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Submission Week:	Week 6 Second Block

This module is presented on NQF level 6.

5% will be deducted from the student's project mark for each calendar day the project is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual project.

This project contributes 40% towards the final mark.

[1] Under no circumstances will projects be accepted for marking after the projects of other students have been marked and returned to the students.

2. Instructions to Students

Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted projects.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your projects.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing (See the "instructions to students" book activity before this activity where the application of negative marking is explained).
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

3. Section A

Section A

Learning Objective

Demonstrate an understanding of Python programming concepts.

Use Python to communicate with individual network devices.

Use Python to communicate with multiple devices.

Demonstrate knowledge in relation to the design of HTTP protocols using Python tools.

Apply the concept of email communication in Python.

Use test methodologies for the creation of quality applications.

Project Topic

Design of a Shopping Store

Scope

Block 1 and 2, Week 1 – 7

Technical Aspects

Python 3

3.1. Question 1

Question 1 60 Marks

Study the scenario and complete the question(s) that follow:

Aircraft Network Monitoring System

You are a Network Systems Engineer at StarLink Airlines. You must design a distributed system to monitor aircraft operations using TCP and HTTP communication, database storage, multi-client handling, real-time monitoring, and security.

The system must:

- Collect aircraft data (e.g., boarding, fuel, autopilot, cabin pressure).
- Transmit this data using network communication (TCP and HTTP).
- Store and retrieve data from a central database.
- Support multiple users accessing the system at the same time.
- Continuously monitor flight conditions and trigger alerts when unsafe situations occur.
- Ensure secure communication between systems.

The system simulates interaction between:

- Ground staff systems (clients)
- A central server
- A flight data generator
- A monitoring and alert system

Design and implement a real-time, network-based monitoring system using Python that demonstrates:

- Client-server communication
- API development
- Database integration
- Concurrency
- Monitoring and security

Source: Nduwamungu, M. (2026).

1.1 TCP Communication Layer

You are required to design and implement an HTTP server-based system that provides access to pre-departure aircraft information for Aero Airlines.

The system must simulate a centralised API service used by ground control systems and external clients to retrieve flight readiness data.

<u>You must</u>:

1. Develop an HTTP server using Python that:

- Listens for incoming client requests
- Processes HTTP GET requests
- Returns aircraft data in JSON format

2. The server must read flight data from a structured file (e.g., .txt, .csv, or .json) containing pre-departure information.

3. The server must expose an endpoint such as: GET /flight

4. Upon receiving a request, the server must:

- Read the file contents
- Convert the data into a structured JSON response
- Return the response with the appropriate HTTP status code (200 OK)

Sample Input Data (File Content)

Aircraft Status: Ready for departure

Passenger Boarding Number: 128

Fueling: Completed

Door State: Closed

Push Back Time: 13:45

(10 Marks)

1.2 Client Application for Data Retrieval

Design and implement a client-side application that connects to the central monitoring server and retrieves Pre-Departure aircraft information.

Requirements:

- Establish communication with the server using an appropriate protocol (TCP or HTTP)
- Request and receive structured aircraft data
- Display the retrieved data in a clear and readable format
- Handle connection or data errors appropriately

Expected Output: Successfully retrieved and displayed pre-departure data

(10 Marks)

1.3 HTTP-Based Data Update System

Develop an application used by ground staff at the departure gate to update Pre-Departure aircraft information via HTTP requests.

Requirements:

- Implement an HTTP-based interaction between the client application and the server
- Allow updates to key aircraft attributes such as: Aircraft status Passenger count Fueling status
- Ensure that the server correctly processes and stores the updated data
- Use appropriate HTTP methods (e.g., POST/PUT).

Expected Output: Successful submission and confirmation of updated aircraft data

(10 Marks)

1.4 HTTP Server with Database Integration

Design and implement an HTTP server system that retrieves In-Flight Events data from a relational database and serves it to clients in JSON format.

Requirements:

- Connect to a MySQL or PostgreSQL database
- Retrieve stored in-flight event data (e.g., autopilot status, cabin pressure, Wi-Fi usage)
- Expose an HTTP endpoint (e.g., /events)
- Return data as a properly structured JSON response
- Implement basic error handling for database and server operations

Expected Output: JSON response containing in-flight event data accessible via an HTTP

(10 Marks)

1.5 Client Application for API Consumption

Develop a client application that consumes the HTTP API created in Question 1.4 and retrieves In-Flight Events information.

Requirements:

- Send HTTP requests to the server endpoint
- Parse the JSON response
- Display the information in a structured format
- Handle invalid responses or connection failures

Expected Output: Successfully retrieved and displayed in-flight event data.

(10 Marks)

1.6 Automated Real-Time Data Update System

Design and implement an automated application that simulates real-time In-Flight Events updates by transmitting data to the server at regular intervals.

Requirements:

- Generate simulated flight data (e.g., cabin pressure, autopilot status)
- Send updates to the server every 5 seconds
- Ensure that the transmitted data is stored in the database
- Maintain continuous operation over time

Expected Output: Continuous data updates reflected in the system database.

(10 Marks)

End of Question 1

3.2. Question 2

Question 2 20 Marks

Study the scenario and complete the question(s) that follow:

Real-Time Flight Safety Monitoring System

Aircraft data collection is essential for ensuring aviation safety and efficient operations. Critical information such as position, altitude, speed, and engine status enables Air Traffic Control (ATC) and airline operations centres to monitor flights in real time. This allows for early detection of anomalies and supports immediate decision-making when deviations or technical issues occur.

In this project, you are required to design and implement a system that simulates the collection and transmission of flight data to a central server. Using Python, you will develop a network-based solution that communicates via TCP sockets or HTTP, ensuring that data is structured and transmitted reliably between components. The system must also include a database component where incoming flight data is stored and can be retrieved for analysis. This supports both real-time monitoring and historical tracking, which are important for identifying patterns and improving operational efficiency.

Finally, you will implement a monitoring mechanism that evaluates incoming data against predefined thresholds to detect abnormal conditions. When such conditions occur, the system should generate alerts, demonstrating how real-time data collection contributes to proactive risk management and improved aviation safety.

Source: Nduwamungu, M. (2026).

2.1 Aero Airlines requires a real-time safety monitoring module as part of its Aircraft Network Monitoring System to ensure that critical in-flight parameters remain within safe operational limits. You are required to design and implement a Python-based monitoring application that continuously evaluates incoming flight data and generates alerts when abnormal conditions are detected.

a. Define Safety Parameters

Establish appropriate safety thresholds for the following in-flight parameters:

- Cabin pressure
- Autopilot status
- Wi-Fi usage

Each parameter must include:

- Acceptable operating range or condition
- Justification for the chosen threshold

(5 Marks)

b. Data Acquisition

Develop a mechanism to retrieve the latest in-flight event data, either by:

- Reading from a database, OR
- Consuming data from an HTTP API endpoint, OR
- Receiving data from a simulated data stream

The system must handle continuously updating data.

(5 Marks)

c. Monitoring Logic Implementation

Implement a monitoring function that:

- Continuously evaluates incoming data (real-time or near real-time loop)
- Compares values against defined safety thresholds
- Detects abnormal or unsafe conditions

(5 Marks)

d. Alert Generation

When unsafe conditions are detected, the system must:

- Generate clear and descriptive alert messages

And indicate:

- Parameter affected
- Current value
- Expected safe range

(3 Marks)

e. System Efficiency and Reliability

Ensure that your solution:

- Runs continuously without crashing
- Handles invalid or missing data gracefully

(2 Marks)

Expected Outcome

Your solution should simulate a real-world flight monitoring system, demonstrating:

- Continuous data processing
- Decision-making based on thresholds
- Integration with networked or stored data
- Clear system feedback (alerts)

End of Question 2

3.3. Question 3

Question 3 20 Marks

Study the scenario and complete the question(s) that follow:

Real-Time Flight Safety Monitoring and Alert System

Network communication plays a critical role in aviation by enabling real-time data exchange between aircraft systems, ground control, and monitoring platforms. This constant flow of information ensures that key operational data such as flight status, environmental conditions, and system performance is shared accurately and promptly. As a result, communication systems directly contribute to safer and more efficient flight operations.

In modern aviation systems like the Aero Airlines monitoring platform, different communication protocols are used depending on the task. Low-level communication is handled using TCP, which provides reliable and continuous data transmission between systems. This is essential for sending critical operational data where accuracy and delivery assurance are required.

On the other hand, HTTP is used for higher-level communication, particularly for accessing and managing data through APIs. It allows different components of the system, such as user interfaces and monitoring dashboards, to request and retrieve structured information in a standardised format like JSON. This makes the system more flexible and easier to integrate across multiple platforms.

Together, TCP and HTTP complement each other within the system. TCP ensures stable and secure data transmission at the network level, while HTTP enables efficient data access and interaction at the application level. Their combined use supports a scalable, reliable, and real-time aircraft monitoring system.

Source: Nduwamungu, M. (2026).

You are required to design and implement a server-side monitoring module for the Aero Airlines system that continuously evaluates in-flight operational data and triggers alerts when safety thresholds are violated.

Your program must:

1. Data Processing

Retrieve or receive in-flight event data from the system (e.g., database, API, or incoming stream), including:

- Cabin pressure
- Autopilot status
- Wi-Fi usage

2. Safety Threshold Evaluation

Define and implement appropriate safety thresholds, for example:

- Cabin pressure must remain within a safe operational range
- Autopilot must remain engaged during flight
- Abnormal system usage patterns must be detected

3. Alert Generation (Email Notification)

When a safety violation is detected, the system must:

- Automatically generate an email alert containing:
    - The parameter that triggered the alert
    - The recorded value
    - Timestamp of the event
    - A brief description of the issue
- Send the alert to a monitoring service email address

4. Integration with Server Architecture

Your solution must:

- Be implemented as part of the central monitoring server, and
- Demonstrate how it integrates with the overall system (data source + alert mechanism)

Expected Output Example

```
ALERT: Cabin Pressure Out of Range

Value: 8.9 PSI

Time: 14:32:10

Action: Email notification sent to monitoring@aeroairlines.com
```

(20 Marks)

End of Question 3